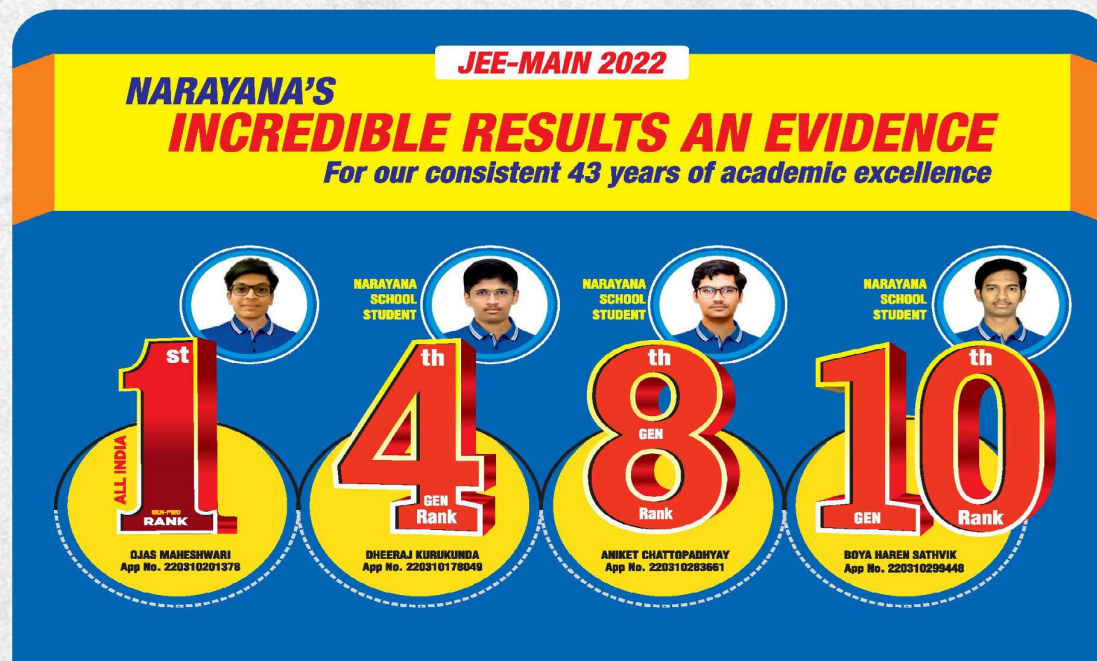




NARAYANA
IIT-JEE ACADEMY - INDIA

NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
EXCELLENCE IS OUR TRADITION 3402 RANKS OVERALL!



CHEMISTRY

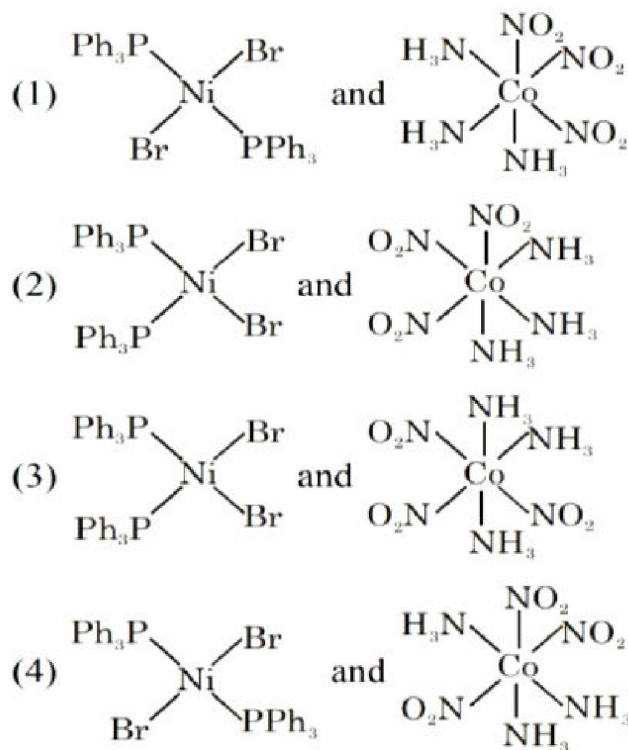
INORGANIC CHEMISTRY PART-II(SR) JEE ADVANCED IMPORTANT QUESTIONS

(Single Correct Type)

- According to molecular theory, the species among the following that does not exist is
A) He_2^+ B) He_2^- C) Be_2 D) O_2^{2-}
- The correct statement about B_2H_6 is
A) Terminal B-H bonds have less p-character when compared to bridging bonds
B) The two B-HB bonds are not of same length
C) All B-H-B angles are of 120°
D) Its fragment, BH_3 , behaves as a Lewis base
- The correct shape and I-I-I bond angles respectively in I_3^- ion are
A) Distorted trigonal planar; 135° and 90°
B) T-shaped; 180° and 90°
C) Trigonal planar; 120°
D) Linear; 180°
- The correct order of bond dissociation enthalpy of halogens is
A) $Cl_2 > F_2 > Br_2 > I_2$ B) $I_2 > Br_2 > Cl_2 > F_2$
C) $Cl_2 > Br_2 > F_2 > I_2$ D) $F_2 > Cl_2 > Br_2 > I_2$
- The oxidation states of 'P' in $H_4P_2O_7$, $H_4P_2O_5$ and $H_4P_2O_6$, respectively, are
A) 7, 5 and 6 B) 5, 4 and 3 C) 5, 3 and 4 D) 6, 4 and 5
- The INCORRECT statement regarding the structure of C_{60} is
A) The six-membered rings are fused to both six and five-membered rings
B) Each carbon atom forms three sigma bonds
C) The five-membered rings are fused only to six-membered rings
D) It contains 12 six-membered rings and 24 five-membered rings
- The calculated magnetic moments (spin only value) for species $[FeCl_4]^{2-}$, $[Co(C_2O_4)_3]^{3-}$ and MnO_4^{2-} respectively are
A) 5.82, 0 AND 0 BM B) 4.90, 0 and 1.73 BM
C) 5.92, 4.90 and 0 BM D) 4.90, 0 and 2.83 BM
- In which of the following order the given complex ions are arranged correctly with respect to their decreasing spin only magnetic moment?
i) $[FeF_6]^{2-}$ ii) $[Co(NH_3)_6]^{3+}$ iii) $[NiCl_4]^{2-}$ iv) $[Cu(NH_3)_4]^{2+}$
A) (i) > (iii) > (iv) > (ii) B) (ii) > (iii) > (i) > (iv)
C) (iii) > (iv) > (ii) > (i) D) (ii) > (i) > (iii) > (iv)
- Arrange the following metal complex/compounds in the increasing order of spin only magnetic moment. Presume all the three, high spin system.
(Atomic numbers $Ce = 58, Gd = 64$ and $Eu = 63$)
1) $(NH_4)_2[Ce(NO_3)_6]$ 2) $Gd(NO_3)_3$ and 3) $Eu(NO_3)_3$

- A) $(2) < (1) < (3)$ B) $(3) < (1) < (2)$ C) $(1) > (2) > (3)$ D) $(1) < (3) < (2)$

10. The correct structures of trans - $[NiBr_2(PPh_3)_2]$ and meridional - $[Co(NH_3)_3(NO_2)_3]$, respectively are



11. The correct order of intensity of colors of the compounds is

- A) $[Ni(CN)_4]^{2-} > [NiCl_4]^{2-} > [Ni(H_2O)_6]^{2+}$
 B) $[Ni(H_2O)_6]^{2+} > [NiCl_4]^{2-} > [Ni(CN)_4]^{2-}$
 C) $[NiCl_4]^{2+} > [Ni(H_2O)_6]^{2+} > [Ni(CN)_4]^{2-}$
 D) $[NiCl_4]^{2-} > [Ni(CN)_4]^{2-} > [Ni(H_2O)_6]^{2+}$

12. Spin only magnetic moment of an octahedral complex of Fe^{2+} in the presence of a strong field ligand in BM is

- A) 4.89 B) 2.82 C) 0 D) 3.46

13. Which one of the following metal complexes is most stable?

- A) $[Co(en)(NH_3)_4]Cl_2$ B) $[Co(en)_3]Cl_2$
 C) $[Co(en)_2(NH_3)_2]Cl_2$ D) $[Co(NH_3)_6]Cl_2$

14. The number of geometrical isomers found in the metal complexes

$[PtCl_2(NH_3)_2]$, $[Ni(CO)_4]$, $[Ru(H_2O)_3Cl_3]$ and $[CoCl_2(NH_3)_4]^+$ respectively, are

- A) 1,1,1,1 B) 2,1,2,2 C) 2,0,2,2 D) 2,1,2,1

15. Indicate the complex/complex ion which did not show any geometric at isomerism
 A) $[CoCl_2(en)_2]$ B) $[Co(CN)_5(NC)]^{3-}$ C) $[Co(NH_3)_3(NO_2)_3]$ D) $[Co(NH_3)_4Cl_2]^+$
16. What is the correct order of the following elements with respect to their density?
 A) $Cr < Zn < Co < Cu < Fe$ B) $Zn < Cu < Co < Fe < Cr$
 C) $Zn < Cr < Fe < Co < Cu$ D) $Cr < Fe < Co < Cu < Zn$
17. The incorrect statement among the following is
 A) $VOSO_4$ is a reducing agent B) Cr_2O_3 is an amphoteric oxide
 C) RuO_4 is an oxidizing agent D) Red colour of ruby is due to the presence of Co^{3+}
18. Fex_2 and Fey_3 are known when x and y are
 A) $x = F, Cl, Br, I$ and $y = F, Cl, Br$ B) $x = F, Cl, Br$ and $y = F, Cl, Br, I$
 C) $x = Cl, Br, I$ and $y = F, Cl, Br, I$ D) $x = F, Cl, Br, I$ and $y = F, Cl, Br, I$
19. The common positive oxidation states for an element with atomic number 24, are
 A) +2 to +6 B) +1 and +3 to +6 C) +1 and +3 D) +1 to +6
20. Cu^{2+} salt reacts with potassium iodide to give
 A) Cu_2I_2 B) Cu_2I_3 C) CuI D) $Cu(I_3)_2$
21. The set having ions which are coloured and paramagnetic both is
 A) Cu^{2+}, Cr^{3+}, Sc^{+} B) $Cu^{2+}, Zn^{2+}, Mn^{4+}$ C) Sc^{3+}, V^{5+}, Ti^{4+} D) $Ni^{2+}, Mn^{7+}, Hg^{2+}$
22. Which one of the following metals forms interstitial hydride easily?
 A) Cr B) Fe C) Mn D) Co
23. Which one of the following when dissolved in water gives coloured solution in nitrogen atmosphere?
 A) $CuCl_2$ B) $AgCl$ C) $ZnCl_2$ D) Cu_2Cl_2
24. The nature of oxides V_2O_3 and CrO is indexed as 'X' and 'Y' type respectively. The correct set of X and Y is
 A) X = basic Y = amphoteric
 B) X = amphoteric Y = basic
 C) X = acidic Y = acidic
 D) X = basic Y = basic
25. The addition of dilute $NaOH$ to Cr^{3+} salt solution will give
 A) A solution of $[Cr(OH)_4]^-$ B) Precipitate of $Cr_2O_3(H_2O)_n$
 C) Precipitate of $[Cr(OH)_6]^{3-}$ D) Precipitate of $Cr(OH)_3$
26. Potassium permanganate on heating at 513 K gives a product which is
 A) Paramagnetic and colourless B) Diamagnetic and green
 C) Diamagnetic and colourless D) Paramagnetic and green
27. In the structure of the dichromate ion, there is a
 A) Linear symmetrical $Cr-O-Cr$ bond B) Non-linear symmetrical $Cr-O-Cr$ bond
 C) Linear unsymmetrical $Cr-O-Cr$ bond D) Non-linear unsymmetrical $Cr-O-Cr$ bond

28. In which one of the following sets all species show disproportionation reaction?
 A) ClO_2^- , F_2 , MnO_4^{2-} and $\text{Cr}_2\text{O}_7^{2-}$ B) $\text{Cr}_2\text{O}_7^{2-}$, MnO_4^{2-} , ClO_2^- and Cl_2
 C) MnO_4^{2-} , ClO_2^- , Cl_2 and Mn^{3+} D) ClO_4^- , MnO_4^{2-} , ClO_2^- and F_2
29. In the given chemical reactions, colors of the Fe^{2+} and Fe^{3+} ions, are respectively
 $5\text{Fe}^{2+} + \text{MnO}_4^- + 8\text{H}^+ \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{3+}$
 A) Yellow, Orange B) Yellow, Green C) Green, Orange D) Green, Yellow
30. Identity the element for which electronic configuration in +3 oxidation state is $[\text{Ar}]3d^5$
 A) Ru B) Mn C) Co D) Fe
31. Which of the following ore is concentrated using group 1 cyanide salt?
 A) Sphalerite B) Calamine C) Siderite D) Malachite
32. Al_2O_3 was leached with alkali to get X. The solution of X on passing of gas Y, forms Z. X, Y and Z respectively are
 A) $\text{X} = \text{Na}[\text{Al}(\text{OH})_4]$, $\text{Y} = \text{SO}_2$, $\text{Z} = \text{Al}_2\text{O}_3$ B) $\text{X} = \text{Na}[\text{Al}(\text{OH})_4]$, $\text{Y} = \text{CO}_2$, $\text{Z} = \text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
 C) $\text{X} = \text{Al}(\text{OH})_3$, $\text{Y} = \text{CO}_2$, $\text{Z} = \text{Al}_2\text{O}_3$ D) $\text{X} = \text{Al}(\text{OH})_3$, $\text{Y} = \text{SO}_2$, $\text{Z} = \text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
33. The major components in "Gun Metal" are
 A) Cu, Zn and Ni B) Cu, Sn and Zn C) Al, Cu, Mg and Mn D) Cu, Ni and Fe
34. Ellingham diagram is a graphical representation of
 A) ΔH vs T B) ΔG vs T C) ΔG vs P D) $(\Delta G - T\Delta S)$ vs T
35. The major components of German Silver are
 A) Ge, Cu and Ag B) Zn, Ni and Ag C) Cu, Zn and Ni D) Cu, Zn and Ag
36. The method used for the purification of Indium is
 A) Van arkel method B) Liquation C) Zone refining D) Vapour phase refining
37. The process that involves the removal sulphur from the ores is
 A) Smelting B) Roasting
 C) Leaching D) Refining
38. Which of the following reduction reaction CANNOT be carried out with coke?
 A) $\text{Al}_2\text{O}_3 \rightarrow \text{Al}$ B) $\text{ZnO} \rightarrow \text{Zn}$ C) $\text{Fe}_2\text{O}_3 \rightarrow \text{Fe}$ D) $\text{Cu}_2\text{O} \rightarrow \text{Cu}$
39. The chemical that is added to reduce the melting point of the reaction mixture during the extraction of aluminium is
 A) Cryolite B) Bauxite C) Calamine D) Kaolite

40. The metal that can be purified economically by fractional distillation method is
A) Fe B) Zn C) Cu D) Ni
41. Consider two chemical reactions (A) and (B) that take place during metallurgical process
A) $ZnCO_{3(s)} \xrightarrow{\Delta} ZnO_{(s)} + CO_{2(g)}$
B) $2ZnS_{(s)} + 3O_{2(g)} \xrightarrow{\Delta} 2ZnO_{(s)} + 2SO_{2(g)}$
A) (A) is calcination and (B) is roasting
B) Both (A) and (B) are producing same product so both are roasting
C) Both (A) and (B) are producing same product so both are calcination
D) (A) is roasting and (B) is calcination
42. In the leaching of alumina from bauxite, the ore expected to leach out in the process by reacting with $NaOH$ is
A) TiO_2 B) Fe_2O_3 C) ZnO D) SiO_2
43. The statement that is INCORRECT about Ellingham diagram is
A) Provides idea about the reaction rate
B) Provides idea about free energy change
C) Provides idea about changes in the phases during the reaction
D) Provides idea about reduction of metal oxide
44. The addition of silica during the extraction of copper from its sulphide ore
A) Converts copper sulphide into copper silicate
B) Converts iron oxide into iron silicate
C) Reduces copper sulphide into metallic copper
D) Reduces the melting point of the reaction mixture
45. Which refining process is generally used in the purification of low melting metals?
A) Chromatographic method B) Liquation
C) Electrolysis D) Zone refining
46. Calamine and Malachite, respectively, are the ores of
A) Nickel and Aluminium B) Zinc and Copper
C) Copper and Iron D) Aluminium and Zinc

(Multi Correct Type)

47. Which of the following are isostructural pairs?
 A) SO_4^{2-} and CrO_4^{2-} B) $SiCl_4$ and $TiCl_4$ C) NH_3 and NO_3^- D) BCl_3 and $BrCl_3$
48. The green house gas/es is (are)
 A) Carbon dioxide B) Oxygen C) Water vapour D) Methane
49. Which of the following statements(s) is (are) incorrect reason for eutrophication?
 A) excess usage of fertilizers
 B) excess usage of detergents
 C) dense plant population in water bodies
 D) lack of nutrients in water bodies that prevent plant growth
50. The statements that are TRUE:
 A) Methane leads to both global warming and photochemical smog
 B) Methane is generated from paddy fields
 C) Methane is a stronger global warming gas than CO_2
 D) Methane is a part of reducing smog
51. Sulphide ion is soft base and its ores are common for metals
 A) Pb B) Al C) Ag D) Mg

(STATEMENT TYPE)

52. Given below are two statements one is labelled as Assertion A and other is labelled as Reason R
Assertion A: Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond information
Reason R: Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.
 In the light of the above statements, choose the most appropriate answer from the options given below
 A) A is false but R is true
 B) Both A and R are true and R is the correct explanation of A
 C) A is true but R is false
 D) Both A and R are true, but R is NOT the correct explanation of A
53. Given below are two statements one is labelled as assertion A and other is labelled as Reason R
Assertion A: The H-O-H bond angle in water molecule is 104.5° .
Reason R: The lone pair-lone pair repulsion of electrons is higher than the bond pair-bond pair repulsion.

- A) A is false but R is true.
 B) Both A and R are true, but R is NOT the correct explanation of A.
 C) A is true but R is false.
 D) Both A and R are true, and R is the correct explanation of A.
54. Given below are two statements; one is labelled as Assertion A and other is labelled as Reason R.
Assertion A: In $TlCl_3$, isomorphs to CsI_3 , the metal is present in +1 oxidation state.
Reason R: Tl metal has fourteen f electrons in the electronic configuration.
 In the light of the above statements, choose the most appropriate answer from the options given below
 A) A is correct but R is not correct.
 B) Both A and R are true and R is the correct explanation of A.
 C) A is not correct but R is false correct.
 D) Both A and R are true, but R is NOT the correct explanation of A.
55. Given below are two statements one is labelled as Assertion A and other is labelled as Reason R
Assertion A: Lithium halides are somewhat covalent in nature.
Reason R: Lithium process high polarization capability.
 A) A is true but R is false
 B) Both A and R are true, but R is NOT the correct explanation of A
 C) A is false but R is true
 D) Both A and R are true, and R is the correct explanation of A
56. Given below are two statements one is labelled as Assertion A and other is labelled as Reason R
Assertion A: - Barium carbonate is insoluble in water and highly stable.
Reason R: - the thermal stability of the carbonates increases with increasing cationic size.
 A) Both A and R are true and R is the correct explanation of A.
 B) A is true but R is false.
 C) Both A and R are true, but R is NOT the correct explanation of A.
 D) A is false but R is true.
57. Given below are two statements:
Statement I: o-Nitro phenol is steam volatile due to intramolecular hydrogen bonding
 In the light of the above statements, choose the most appropriate answer from the options given below
Statement II: o-Nitrophenol has high melting due to hydrogen bonding.
 In the light of the above statements, choose the most appropriate answer from the options given below
 A) Statement I is false but Statement II is true
 B) Both statement I and Statement II are true
 C) Both statement I and Statement II are false
 D) Statement I is true and Statement II is false
58. Given below are two statements:
Statement I- The identification of Ni^{2+} is carried out by dimethyl glyoxime in the presence of NH_4OH .
Statement II- The dimethyl glyoxime is a bidentate neutral ligand.
 In the light of the above statements, choose the most appropriate answer from the options given below

- A) Statement I is false but Statement II is true
 B) Both statement I and Statement II are false
 C) Statement I is true and Statement II is false
 D) Both statement I and Statement II are true

59. Given below are two statements:

Statement I: $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$ are d^2sp^3 hybridized.

Statement II: $[MnCl_6]^{3-}$ and $[FeF_6]^{3-}$ are paramagnetic and have 4 and 5 unpaired electrons respectively.

In the light of the above statements, choose the most appropriate answer from the options given below

- A) Statement I is correct and Statement II is incorrect
 B) Both statement I and Statement II are false
 C) Statement I is incorrect but Statement II is correct
 D) Both statement I and Statement II are true

60. Given below are two statements

Statement I: Potassium permanganate on heating at 573 K forms potassium manganate.

Statement II: Both potassium permanganate and potassium manganate are tetrahedral and paramagnetic in nature

In the light of the above statements, choose the most appropriate answer from the options given below

- 1) Statement I is true but statement II is false
 2) Both statement I and statement II are true
 3) Statement I is false but statement II is true
 4) Both statement I and statement II are false

61. Given below are two statements:

Statement-I: The value of the parameter "Biochemical Oxygen Demand (BOD)" is important for survival of aquatic life

Statement-II: The optimum value of BOD is 6.5 ppm.

In the light of the above statements, choose the most appropriate answer from the options given below

- A) Statement I is incorrect but Statement II is correct
 B) Both statement I and Statement II are true
 C) Statement I is correct and Statement II is incorrect
 D) Both statement I and Statement II are false

62. Given below are two statements:

Statement-I: An allotrope of oxygen is an important intermediate in the formation of reducing smog.

Statement-II: Gases such as oxides of nitrogen and Sulphur present in troposphere contribute to the formation of photochemical smog

In the light of the above statements, choose the most appropriate answer from the options given below

- A) Both statement I and Statement II are false
 B) Statement I is correct and Statement II is incorrect
 C) Both statement I and Statement II are true
 D) Statement I is incorrect but Statement II is correct

63. Given below are two statements:

Statement-I the pH of rain water is normally ~ 5.6 .

Statement II: - If the pH of rain water drops below 5.6, it is called acid rain.

In the light of the above statements, choose the most appropriate answer from the options given below

- A) Statement I is correct and Statement II is incorrect
- B) Both statement I and Statement II are false
- C) Statement I is incorrect but Statement II is correct
- D) Both statement I and Statement II are true

64. Given below are two statements:

Statement-I: Non-biodegradable wastes are generated by the thermal power plants.

Statements-II: Bio degradable detergents leads to eutrophication

In the light of the above statements, choose the most appropriate answer from the options given below

- A) Both statement I and Statement II are false
- B) Statement I is correct and Statement II is incorrect
- C) Statement I is incorrect but Statement II is correct
- D) Both statement I and Statement II are true

65. Given below are two statements:

Statement I- Chlorofluoro carbons breakdown by radiation in the visible energy region and release chlorine gas in the atmosphere which then reacts with stratospheric ozone

Statement II- Atmospheric ozone reacts with nitric oxide to give nitrogen and oxygen gases, which add to the atmosphere

For the above statements choose the correct answer given below:

- A) Statement I is incorrect but Statement II is correct
- B) Both statement I and Statement II are false
- C) Statement I is correct and Statement II is incorrect
- D) Both statement I and Statement II are true

66. Given below are two statements; one is labelled as assertion A and other is labelled as Reason R

Assertion A: photochemical smog causes cracking of rubber.

Reason R: Presence of ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate in photochemical smog makes it oxidizing.

Choose the most appropriate answer from the options given below:

- A) Both A and R are true but R is NOT the correct explanation of A
- B) A is false but R is true
- C) A is true but R is false
- D) Both A and R are true, and R is the correct explanation of A

67. Given below are two statements:

Statement-I: the choice of reducing agents for metals extraction can be made by using Ellingham diagram, a plot of ΔG vs temperature

Statement-II: The value of ΔS increases from left to right in Ellingham diagram.

In the light of the above statements, choose the most appropriate answer from the options given below

- A) Both statement I and Statement II are true
- B) Statement I is incorrect and Statement II is correct

- C) Both statement I and Statement II are true
 D) Statement I is correct but Statement II is incorrect

68. Given below are two statements:

Statement-I: Sphalerite is a sulphide ore of zinc and copper glance is a sulphide ore of copper.

Statement-II: It is possible to separate two sulphide ores by adjusting proportion of oil to water or by using 'depressants' in a froth floatation method.

Choose the most appropriate answer from the options given below:

- A) Statement-I is true but Statement-II is false.
 B) Both Statement-I and Statement-II are true
 C) Statement-I is false but Statement-II is true.
 D) Both Statement-I and Statement-II are false.

(Matrix Matching Type)

69. Match List-I with List-II

List-I (Molecule)	List-II (Bond order)
a) Ne_2	i) 1
b) N_2	ii) 2
c) F_2	iii) 0
d) O_2	iv) 3

Choose the correct answer from the options given below:

- A) $a-iii, b-iv, c-i, d-ii$
 B) $a-i, b-ii, c-iii, d-iv$
 C) $a-ii, b-i, c-iv, d-iii$
 D) $a-iv, b-iii, c-ii, d-i$

70. Match List-I with List-II

List-I (Name of oxo acid)	List-II (Oxidation state of 'P')
a) Hypophosphorous acid	i) + 5
b) Orthophosphoric acid	ii) + 4
c) Hypophosphoric acid	iii) + 3
d) Orthophosphorous acid	iv) + 2
	v) + 1

Choose the correct answer from the options given below:

- A) $a-v, b-i, c-ii, d-iii$
 B) $a-iv, b-i, c-ii, d-iii$
 C) $a-iv, b-v, c-ii, d-iii$
 D) $a-v, b-iv, c-ii, d-iii$

71. Match List-I with List-II

List-I (Species)	List-II (Hybrid Orbitals)
a) SF_4	i) sp^3d^2
b) IF_5	ii) d^2sp^3
c) NO_2^+	iii) sp^3d
d) NH_4^+	iv) sp^3
	v) sp

Choose the correct answer from the options given below:

A) $a-i, b-ii, c-v, d-iii$ B) $a-ii, b-i, c-iv, d-v$ C) $a-iii, b-i, c-v, d-iv$ D) $a-iv, b-iii, c-ii, d-v$

72. Match List-I with List-II

List-I (Species)	List-II (Number of lone pairs of electrons on the central atom)
a) XeF_2	i) 0
b) XeO_2F_2	ii) 1
c) XeO_3F_2	iii) 2
d) XeF_4	iv) 3

Choose the correct answer from the options given below:

A) $a-iv, b-i, c-ii, d-iii$ B) $a-iii, b-iv, c-ii, d-i$ C) $a-iii, b-ii, c-iv, d-i$ D) $a-iv, b-ii, c-i, d-iii$

73. Match List-I with List-II

List-I	List-II
a) $[Co(NH_3)_6][Cr(CN)_6]$	i) Linkage isomerism
b) $[Co(NH_3)_3(NO_2)_3]$	ii) Solvate isomerism
c) $[Cr(H_2O)_6]Cl_3$	iii) Co-ordination isomerism
d) $cis-[CrCl_2(ox)_2]^{3-}$	iv) Optical isomerism

Choose the correct answer from the options given below:

A) $a-iii, b-i, c-ii, d-iv$ B) $a-iv, b-ii, c-iii, d-i$ C) $a-ii, b-i, c-iii, d-iv$ D) $a-i, b-ii, c-iii, d-iv$

74. Match List-I with List-II

List-I	List-II
a) Chlorophyll	i) Ruthenium
b) Vitamin- B_{12}	ii) Platinum
c) Anticancer drug	iii) Cobalt
d) Grubbs catalyst	iv) Magnesium

Choose the most appropriate answer from the options given below:

A) $a-iii, b-ii, c-iv, d-i$ B) $a-iv, b-iii, c-ii, d-i$ C) $a-iv, b-iii, c-i, d-ii$ D) $a-iv, b-ii, c-iii, d-i$

75. Match List-I with List-II

List-I (Process)	List-II (Catalyst)
a) Deacon's process	i) ZSM-5
b) Contact process	ii) $CuCl_2$
c) Cracking of hydrocarbons	iii) Particle 'Ni'
d) Hydrogenation of vegetable oils	iv) V_2O_5

Choose the most appropriate answer from the options given below:

A) $a-iii, b-iv, c-i, d-iii$ B) $a-i, b-iii, c-ii, d-iv$ C) $a-iii, b-i, c-iv, d-ii$ D) $a-iv, b-ii, c-i, d-iii$

76. Match List-I with List-II

List-I (Metal)	List-II (Ores)
a) Aluminium	i) Siderite
b) Iron	ii) Calamine
c) Copper	iii) Kaolinite
d) Zinc	iv) Malachite

Choose the most appropriate answer from the options given below:

A) $a-iv, b-iii, c-ii, d-i$ B) $a-ii, b-iv, c-i, d-iii$ C) $a-i, b-ii, c-iii, d-iv$ D) $a-iii, b-i, c-iv, d-ii$

77. Match List-I with List-II

List-I	List-II
a) Kernite	i) Tin
b) Cassiterite	ii) Boron
c) Calamine	iii) Fluorine
d) Cryolite	iv) Zinc

Choose the most appropriate answer from the options given below:

A) $a-i, b-iii, c-iv, d-ii$ B) $a-ii, b-i, c-iv, d-iii$ C) $a-ii, b-iv, c-i, d-iii$ D) $a-iii, b-i, c-ii, d-iv$

78. Match List-I with List-II

List-I	List-II
a) Siderite	i) Cu
b) Calamine	ii) Ca
c) Malachite	iii) Fe
d) Cryolite	iv) Al
	v) Zn

Choose the most appropriate answer from the options given below:

A) $a-iii, b-i, c-v, d-ii$ B) $a-i, b-ii, c-v, d-iii$ C) $a-iii, b-v, c-i, d-iv$ D) $a-i, b-ii, c-iii, d-iv$

79. Match List-I with List-II

List-I	List-II
a) Haematite	i) $Al_2O_3.xH_2O$
b) Bauxite	ii) Fe_2O_3
c) Magnetite	iii) $CuCO_3.Cu(OH)_2$
d) Malachite	iv) Fe_3O_4

Choose the most appropriate answer from the options given below:

A) $a-ii, b-iii, c-i, d-iv$ B) $a-iv, b-i, c-ii, d-iii$ C) $a-i, b-iii, c-ii, d-iv$ D) $a-ii, b-i, c-iv, d-iii$

80. Match List-I with List-II

List-I	List-II
a) Mercury	i) Vapour phase refining
b) Copper	ii) Distillation refining
c) Silicon	iii) Electrolytic refining
d) Nickel	iv) Zone refining

Choose the most appropriate answer from the options given below:

A) $a-i, b-iv, c-ii, d-iii$ B) $a-ii, b-iii, c-i, d-iv$ C) $a-ii, b-iii, c-iv, d-i$ D) $a-ii, b-iv, c-iii, d-i$

81. Match List-I with List-II

List-I	List-II
a) Concentration of Ag ore	i) Reverberatory furnace
b) Blast furnace	ii) Pig iron
c) Blister copper	iii) Leaching with dilute NaCN solution
d) Froth floatation method	iv) Sulfide ores

Choose the most appropriate answer from the options given below:

A) $a-iii, b-ii, c-i, d-iv$ B) $a-iii, b-iv, c-i, d-ii$ C) $a-iv, b-i, c-iii, d-ii$ D) $a-iv, b-iii, c-ii, d-i$

82. Match List-I with List-II

List-I (Name of ore/mineral)	List-II (Chemical formula)
a) Calamine	i) ZnS
b) Malachite	ii) $FeCO_3$
c) Siderite	iii) $ZnCO_3$
d) Sphalerite	iv) $CuCO_3 \cdot Cu(OH)_2$

Choose the most appropriate answer from the options given below:

A) $a-iii, b-iv, c-ii, d-i$ B) $a-iii, b-iv, c-i, d-ii$ C) $a-iv, b-iii, c-i, d-ii$ D) $a-iii, b-ii, c-iv, d-i$ **(Integer Type)**83. In gaseous triethyl amine the “ $-C-N-C-$ ” bond angle is _____ degree.84. The difference between bond orders of CO and NO^+ is $\frac{x}{2}$ where $x =$ _____ (Nearest integer)85. The total number of electrons in all bonding molecular orbitals of O_2^{2-} is _____ (Nearest integer)86. AB_3 is an interhalogen T-shaped molecule. The number of lone pairs of electrons on A is _____.87. Number of bridging CO ligands in $[Mn_2(CO)_{10}]$ is _____.88. The number of stereoisomers possible for $[Co(ox)_2(Br)(NH_3)]^{2-}$ is _____ [ox = oxlate]89. The equivalents of ethylene diamine required to replace the neutral ligands from the coordination sphere of the trans-complex of $CoCl_3 \cdot 4NH_3$ is _____. (Nearest Integer)90. $[Ti(H_2O)_6]^{3+}$ absorbs wavelength 498 nm during a $d-d$ transition. The octahedral splitting energy for the above complex is _____ $\times 10^{-19} J$. (Nearest integer) $\times 10^{-19} Jh = 6.626 \times 10^{-34} Js$; $c = 3 \times 10^8 ms^{-1}$ 91. On complete reaction of $FeCl_3$ with oxalic acid in aqueous solution containing KOH , resulted in the

- formation of product A. The secondary valency of Fe in the product A is ____ (Nearest integer).
92. The total number of unpaired electrons present in the complex $K_3[Cr(oxalate)_3]$ is ____.
93. The spin-only magnetic moment value for the complex $[Co(CN)_6]^{4-}$ is ____ BM [At no. of Co = 27]
94. An aqueous solution of $NiCl_2$ was heated with excess sodium cyanide in presence of strong oxidizing agent to form $[Ni(CN)_6]^{2-}$. The total change in number of unpaired electrons on metal centre is ____.
95. The total number of unpaired electrons present in $[Co(NH_3)_6]Cl_2$ and $[Co(NH_3)_6]Cl_3$ is ____.
96. Three moles of AgCl get precipitated when one mole of an octahedral co-ordination compound with empirical formula $CrCl_3 \cdot 3NH_3 \cdot 3H_2O$ reacts with excess of silver nitrate. The number of chloride ions satisfying the secondary valency of the metal ion is ____.
97. The number of geometrical isomers possible in triamminetrinitrocobalt(III) is X and in trioxalatochromate(III) is Y. Then the value of $X + Y$ is ____.
98. 3 moles of metal complex with formula $Co(en)_2Cl_3$ gives 3 moles of silver chloride on treatment with excess of silver nitrate. The secondary valence of Co in the complex is ____ (Nearest integer)
99. 1 mol of an octahedral metal complex with formula $MCl_3 \cdot 2L$ on reaction with excess of $AgNO_3$ gives 1 mol of $AgCl$. The denticity of Ligand L is ____ (Nearest integer)
100. The number of optical isomers possible for $[Cr(C_2O_4)_3]^{3-}$ is ____.
101. The sum of oxidation states of two silver ions in $[Ag(NH_3)_2][Ag(CN)_2]$ complex is ____.





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